

DevOps Lab Task - Program 1: Build a Docker Container using Flask

This document demonstrates how to build a Docker container from a custom Dockerfile using a Flask application. The task involves creating a simple Flask app, writing a Dockerfile, and building the image using Docker CLI commands.

Steps to Build a Flask Docker Container

Step 1: List the project files

```
$ ls
app.py  Dockerfile  requirements.txt
```

Step 2: Build the Docker image with a tag

```
$ sudo docker build -t flask-image:1.0 .
```

This command uses the Dockerfile to create an image named flask-image with version 1.0.

Step 3: Check the created Docker images

```
$ sudo docker images
```

You should see your custom Flask image listed along with other available Docker images.

```
(1RV24MC114_VENKATESHAHD@ kali) ~[~/mca/devops/Lab/Lab1]
$ ls
app.py  Dockerfile  requirements.txt

(1RV24MC114_VENKATESHAHD@ kali) ~[~/mca/devops/Lab/Lab1]
$ sudo docker build -t flask-image:1.0 .
[sudo] password for 1RV24MC114_VENKATESHAHD:
[+] Building 9.7s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 205B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.11-slimsha256:fa9b525a0be0c5ae5e6f2209f4be6fdc5a15a36fed022144d98ac0d08f876d4
=> resolve docker.io/library/python:3.11-slimsha256:fa9b525a0be0c5ae5e6f2209f4be6fdc5a15a36fed022144d98ac0d08f876d4
=> sha256:870925f757415a696459255b0a20b082e914cac42efaff4424b59e21bfff8bd5b 5.48kB / 5.48kB
=> sha256:1ee9c106547f05aa380c4cdec2837c546439943d73d965a3fc49f228dc8be993 1.29MB / 1.29MB
=> sha256:f002d17b63fe84a7f8a66f20cfa63aec4f6cd2a44069f05b6296b0abfcf2a8e1 14.36MB / 14.36MB
=> sha256:658608001a40155a1d3f5aa7f5a10ba02a7d55697301839dc047c9d549b670bc 240B / 240B
=> sha256:fa9b525a0be0c5ae5e6f2209f4be6fdc5a15a36fed022144d98ac0d08f876d4 10.37kB / 10.37kB
=> sha256:b596083aa14d47c78a652138aa9b0807585499d7c7ec343ae378f6c5770822d 1.75kB / 1.75kB
=> extracting sha256:1ee9c106547f05aa380c4cdec2837c546439943d73d965a3fc49f228dc8be993
=> extracting sha256:f002d17b63fe84a7f8a66f20cfa63aec4f6cd2a44069f05b6296b0abfcf2a8e1
=> extracting sha256:658608001a40155a1d3f5aa7f5a10ba02a7d55697301839dc047c9d549b670bc
=> [internal] load build context
=> transferring context: 481B
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> exporting layers
=> writing image sha256:8bcfc0b7f1c84cb8d14291510e08c4e60e438bc26deae667694e6e23b0a94820
=> naming to docker.io/library/flask-image:1.0

(1RV24MC114_VENKATESHAHD@ kali) ~[~/mca/devops/Lab/Lab1]
$ docker images
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied

(1RV24MC114_VENKATESHAHD@ kali) ~[~/mca/devops/Lab/Lab1]
$ sudo docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
node-docker-app      latest             707aeff40b17       Less than a second ago   1.14GB
flask-image          1.0               8bcfc0b7f1c8       10 seconds ago       140MB
flask-image          latest            cdc77d44836d       47 minutes ago       134MB
node                 latest            5cdc8ed4f59b       29 hours ago         1.13GB
wordpress            latest            6aaaaab8dc988      32 hours ago         734MB
nginx                 latest            90086ef6199dc       7 days ago           152MB
alpine               latest            706db57fb206       4 weeks ago          8.32MB
ubuntu               latest            97bed23a3497       5 weeks ago          78.1MB
nimmis/apache-php5   latest            ebf03727818a       17 months ago        448MB
registry              2                 20b2e080310e       2 years ago          25.4MB
mysql/mysql-server   5.6               8c587c89edee       4 years ago          238MB
```